

Recognizing the Use of Steganography in Forensic Evidence (4e)

Digital Forensics, Investigation, and Response, Fourth Edition - Lab 02

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Time on Task:

91 hours, 7 minutes

Progress:

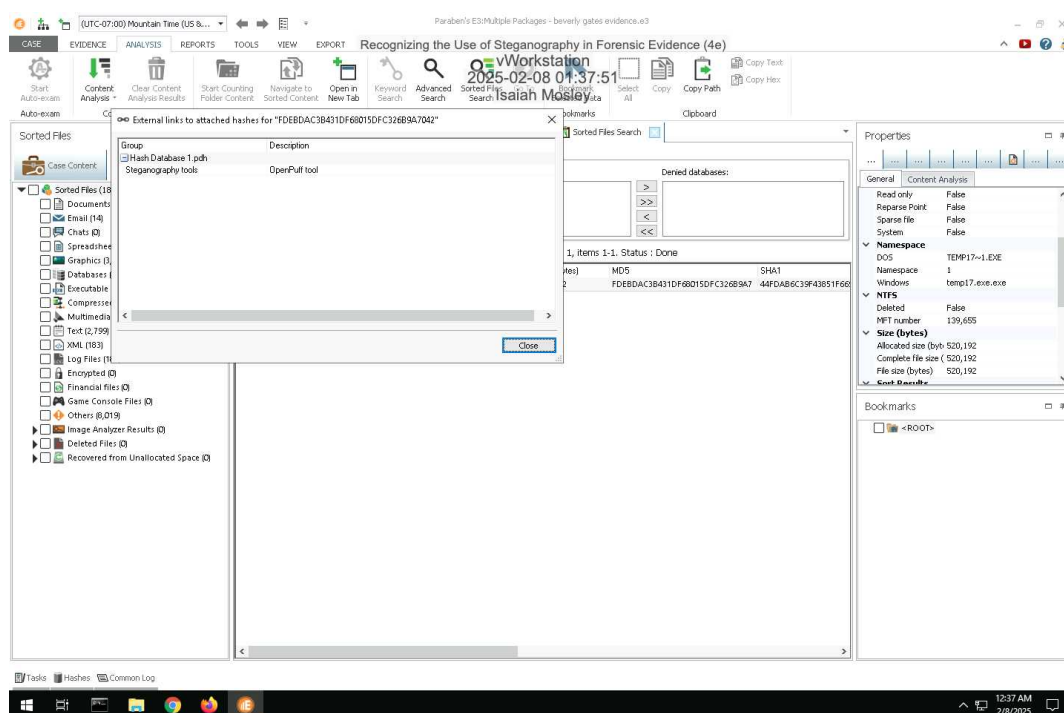
100%

Report Generated: Saturday, February 8, 2025 at 6:08 AM

Section 1: Hands-On Demonstration

Part 1: Detect Steganography Software on a Drive Image

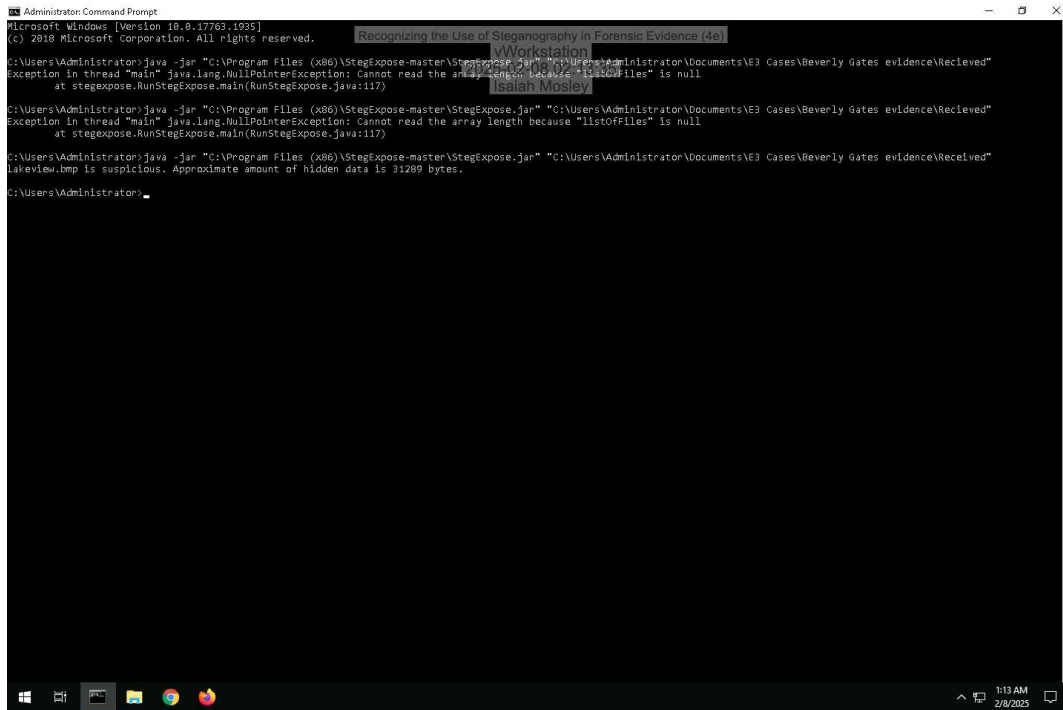
14. Make a screen capture showing the search result and its description.



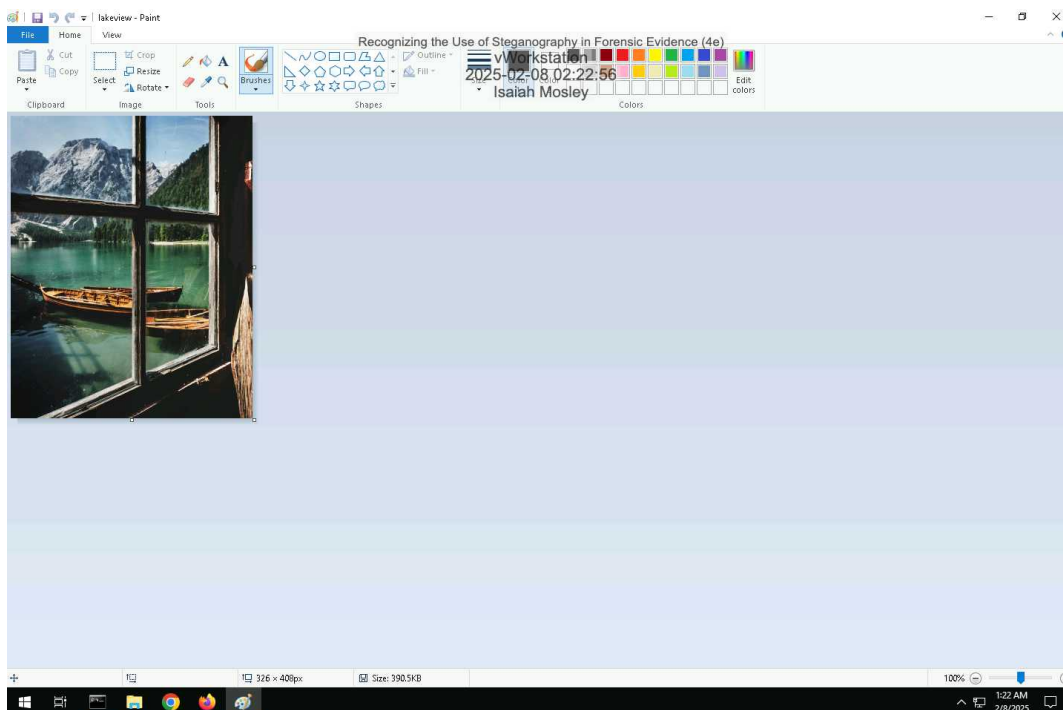
Part 2: Detect Hidden Data in Image Files

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10. **Make a screen capture** showing the **StegExpose** results.



13. **Make a screen capture** showing the **suspicious file** in **Microsoft Paint**.

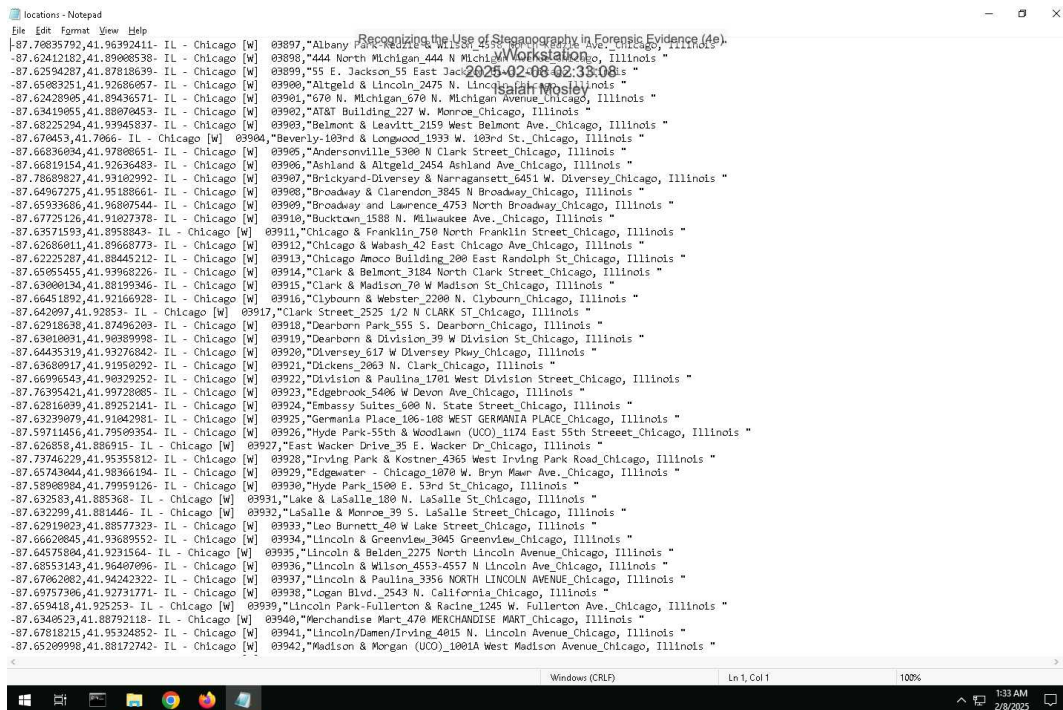


Part 3: Extract Hidden Data from Image Files

2. Record the passphrase saved in the ReadMe file.

landmarks

16. Make a screen capture showing the contents of the file extracted by OpenPuff.



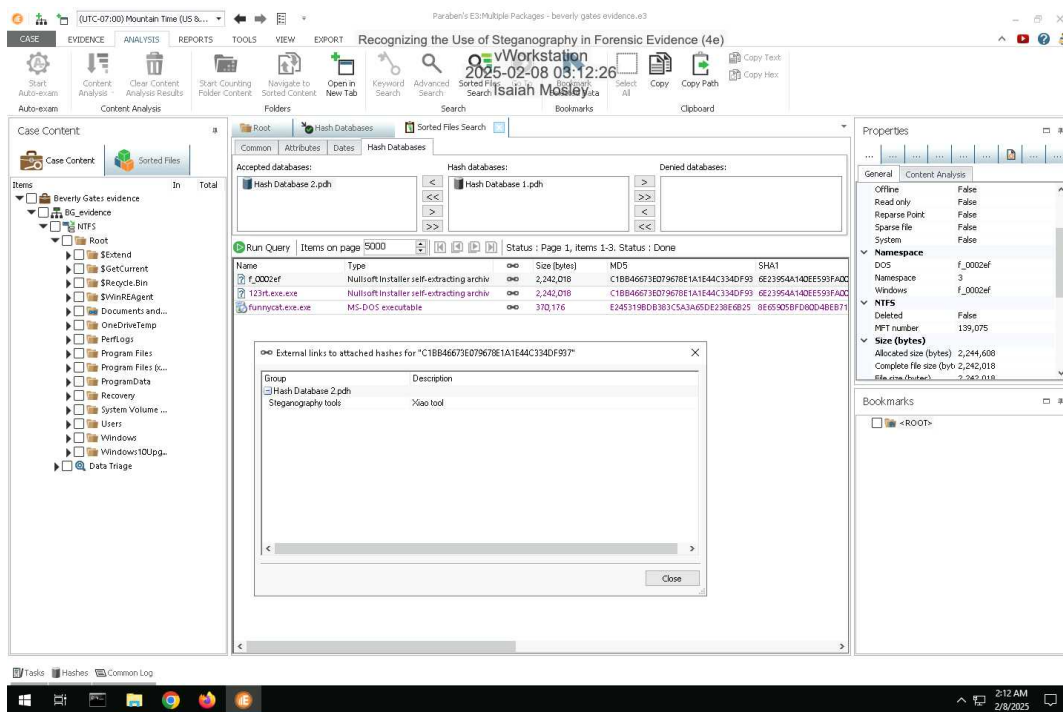
17. Describe the contents of the hidden file. How might it be relevant to the current investigation?

The information in the file contains longitude and latitude coordinates with Chicago, Illinois street addresses. This potentially could be helpful to connect the current investigation to the names and street addresses of who i assume they are selling drugs to.

Section 2: Applied Learning

Part 1: Detect Steganography Software on a Drive Image

5. Make a screen capture showing the search result and its description.



Part 2: Detect Hidden Data in Image and Audio Files

4. Identify the image file with concealed data according to the StegExpose steganalysis tool.

dB9olser.gif

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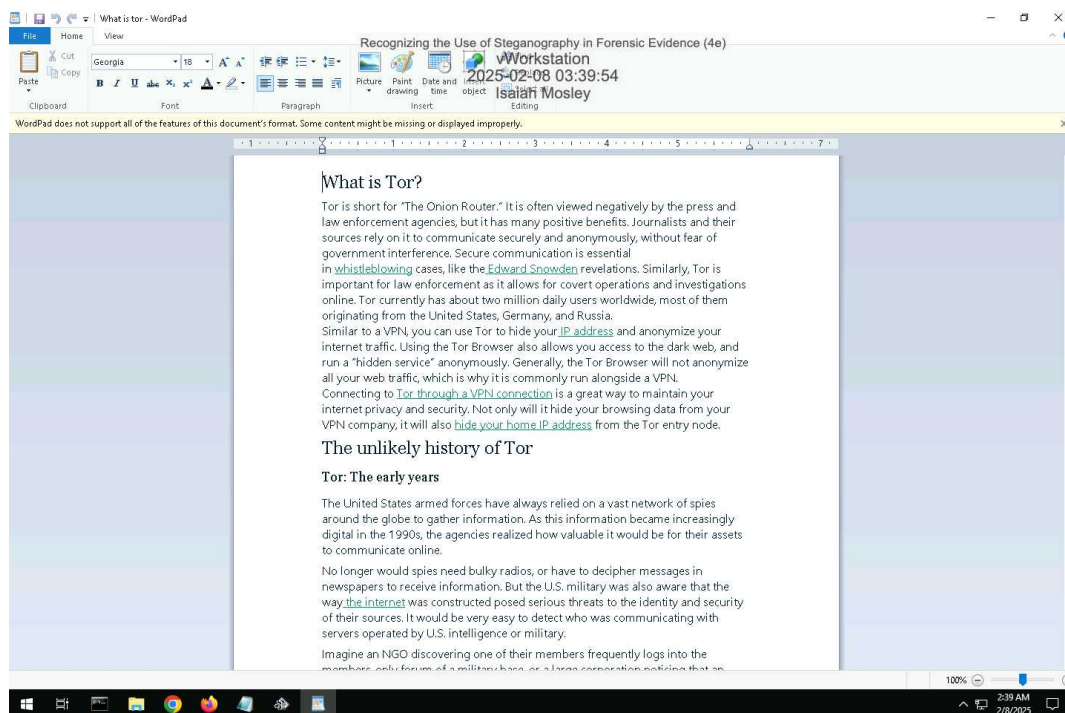
7. Make a screen capture showing the WAV file sizes and hash values in E3.

The screenshot displays the Paraben's E3 Multiple Packages interface. The main window shows a list of files with columns for Name, Type, Size (bytes), MD5, and SHA1. The file 'laugh_x.wav' is highlighted. The Properties pane on the right shows the file's details, including its size (16,304 bytes) and MD5 hash (35D2A83C0A91F8B70B0C).

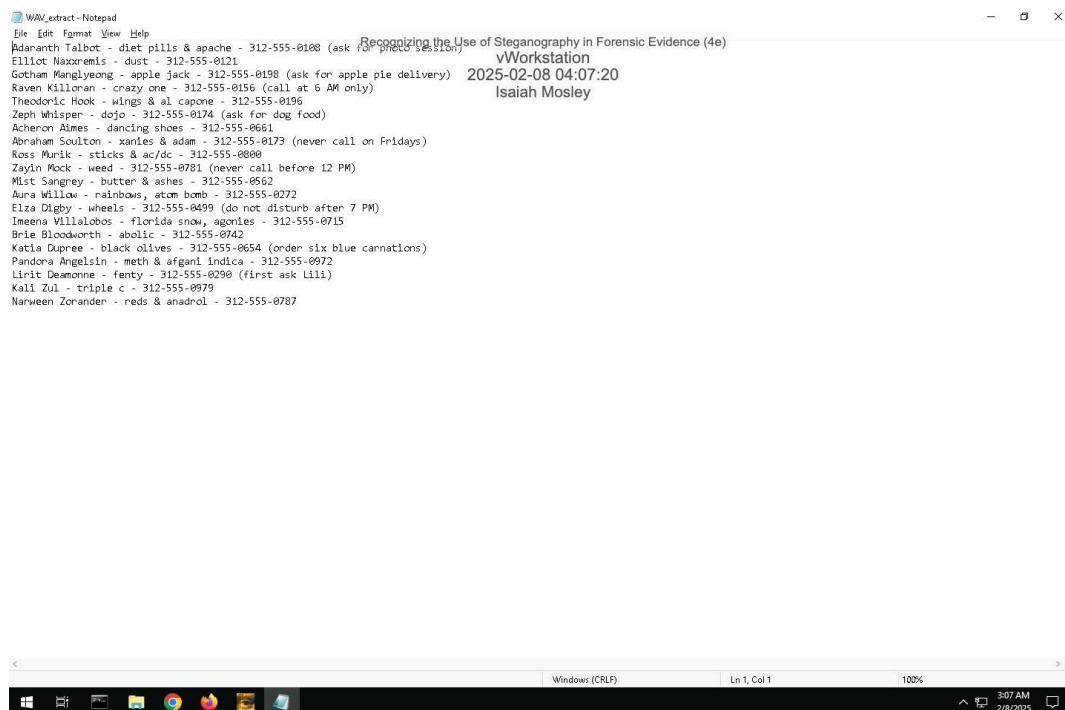
Name	Type	Size (bytes)	MD5	SHA1
laugh_x.wav	Microsoft WAV file format	16,304	35D2A83C0A91F8B70B0C	8BFAA17634D41B
laugh_x.wav	Microsoft WAV file format	16,304	35D2A83C0A91F8B70B0C	8BFAA17634D41B
MANIFEST-000000	MPEG-4 LOAS	50	226F0B1634B18A5051B130F4B3D3D14	56755D205794B3D
MANIFEST-000004	MPEG-4 LOAS	50	031D4D1E2BFA41A96DCB8A21DA92	3CCEB1CB035A0D4
MANIFEST-000014	MPEG-4 LOAS	50	13E4D1C0DF38D4F1592354BBD039A	FFE78F5F6B77F5A6
motorcycle_x.wav	Microsoft WAV file format	53,247	47A41CF04B9CF205D076A56B499C8	8726CEB1F1D05A4C
motorcycle_x1.wav	Microsoft WAV file format	53,248	F93E97018120B0719A3C7B8E8407421	83D4AC4B25CB95
raw_and_check.out.mp3	Audio file with ID3 version 2	37,400	1F16C1A6C0D79921C2D291F2828C457	25C2EF85264AA14

Part 3: Extract Hidden Data from Image and Audio Files

9. Make a screen capture showing the contents of the hidden file extracted by S-Tools.



15. Make a screen capture showing the contents of the hidden file extracted by Xiao.



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16. **Describe** the contents of the two hidden files. How might they be relevant to the current investigation?

The "what is tor" document explains what is tor and how to use it. The WAV file has names, phones numbers and what drugs/items are sold to them on the dark web.

Section 3: Challenge and Analysis

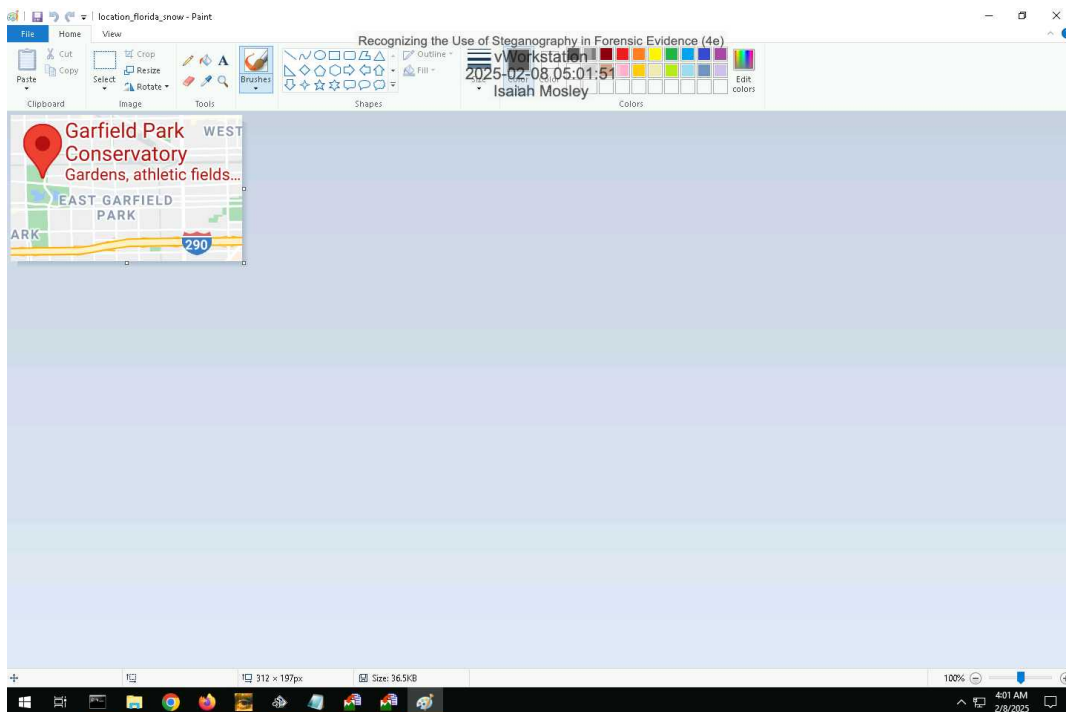
Part 1: Detect More Hidden Data

Record the names of the files that contain concealed data.

chicago.bmp and chicago1.bmp show to be suspicious while using StegExpose on the command prompt.

Part 2: Extract More Hidden Data

Make a screen capture showing the **first file extracted by OpenStego**.



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Make a screen capture showing the second file extracted by OpenStego.

